Sample Frame Construction, Event Study & Spillover Design

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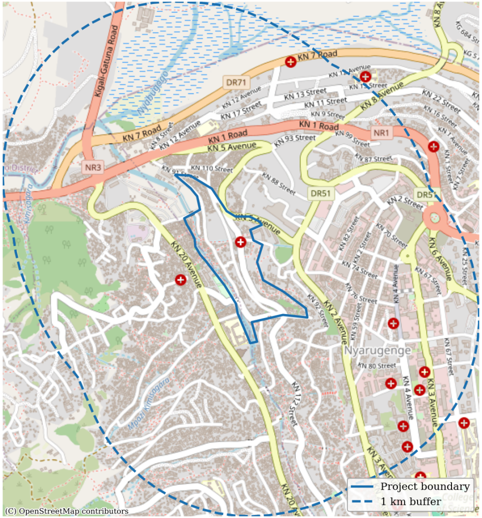
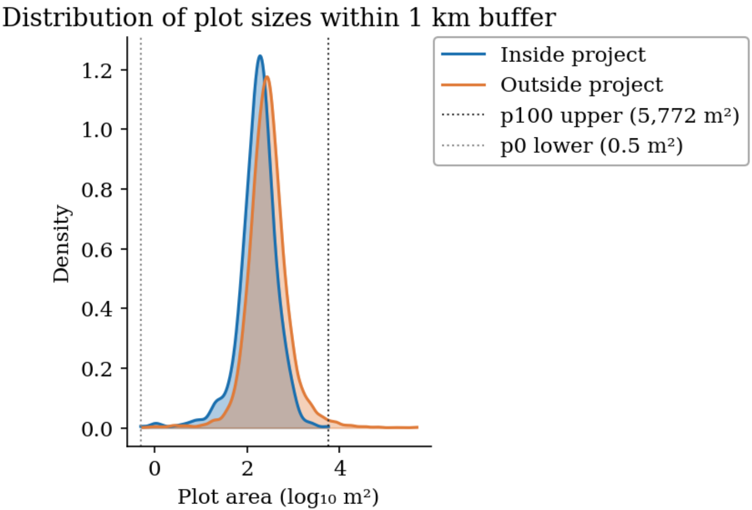
Outline

Context map — OpenStreetMap

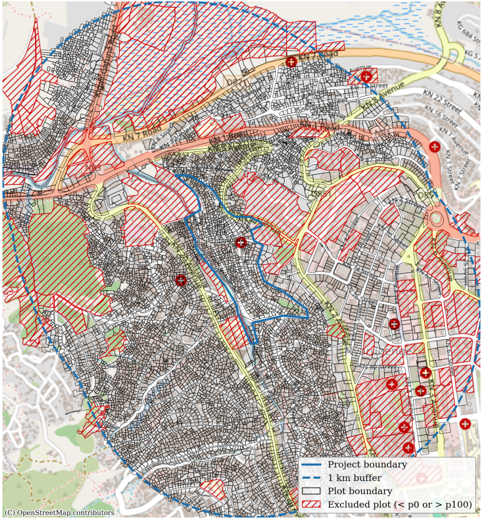


Context map — Satellite imagery

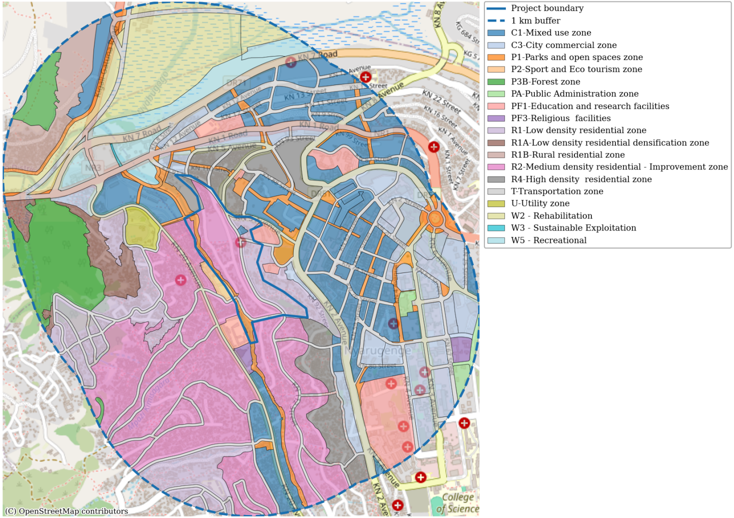
Parcel size distribution



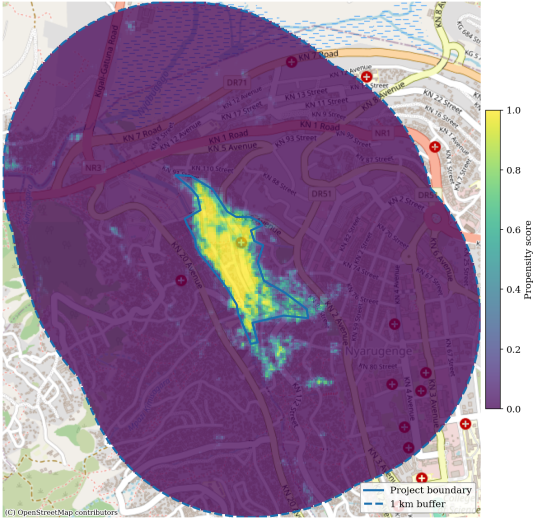
Parcel boundaries within 1 km buffer



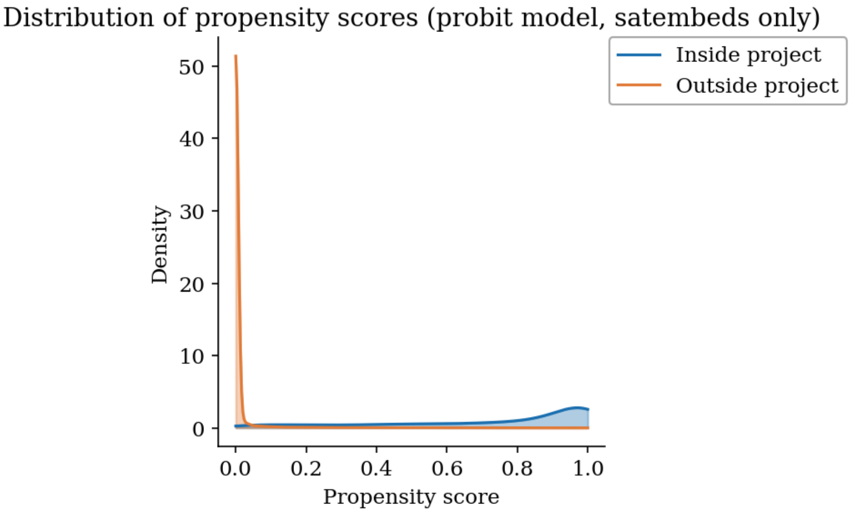
Land-use zoning — Masterplan 2020



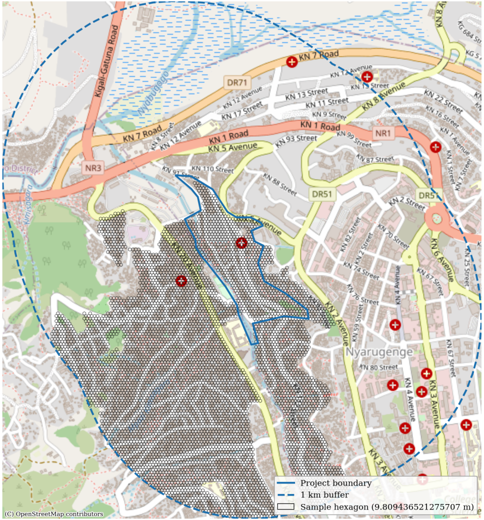
Propensity score map - google satellite embeddings



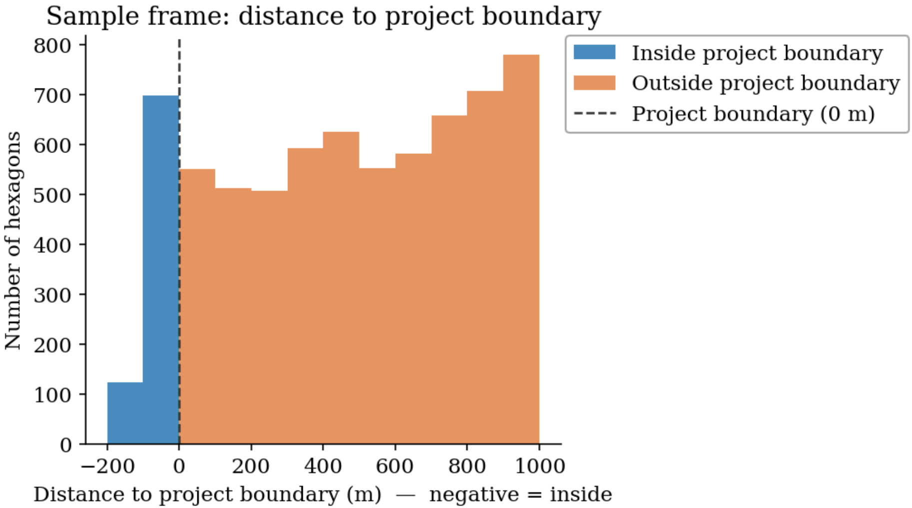
Propensity score distribution



Sample frame — hexagonal grid (250 sqm)



Sample frame — distance to project boundary

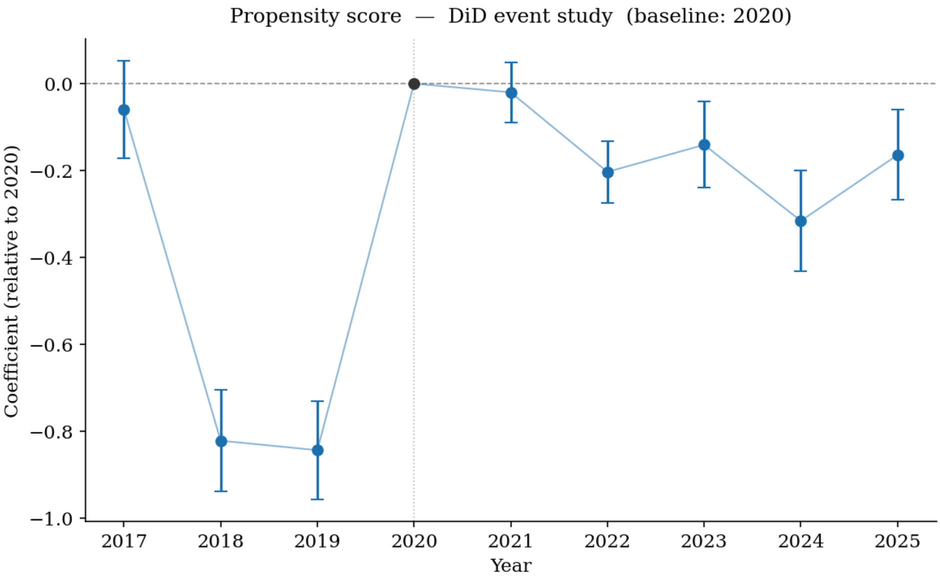


Event study — specification

Estimating equation

$$y_{it} \;=\; \alpha_i + \gamma_t + \sum_{t \neq 2020} \beta_t \cdot \Big(D_i \times \mathbf{1}[t]\Big) + \varepsilon_{it}$$

- i = pixel centroid; t = year (2017–2025)
- y_{it} = propensity score from **Google Satellite Embedding** for pixel i in year t
  - weights estimated using 2020 embeddings, applied to all years
- α_i = pixel fixed effect; γ_t = year fixed effect
- $D_i = 1$ if pixel centroid lies **inside** the project boundary, else 0
- β_t: DiD coefficient for year t relative to baseline 2020 ($\beta_{2020} \equiv 0$)
- Standard errors clustered by 100 m distance bin



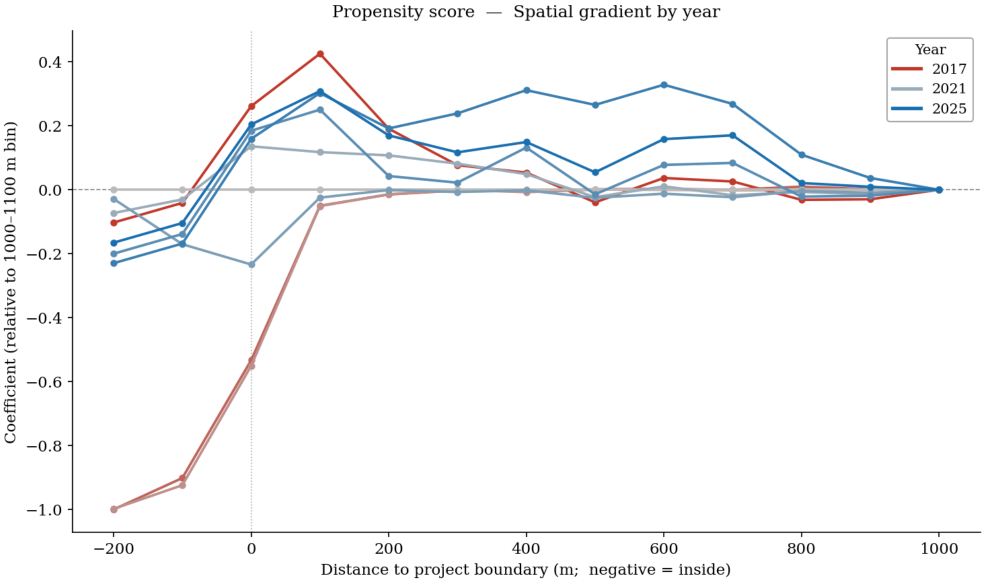
Propensity score — DiD event study (baseline: 2020)
Coefficient (relative to 2020)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

Spatial gradient — specification

Estimating equation

$$y_{it} \;=\; \alpha_i + \sum_{d} \sum_{t \neq 2020} \beta_{d,t} \cdot \Big(\mathbf{1}[\mathrm{bin}_i = d] \times \mathbf{1}[t] \Big) + \varepsilon_{it} \qquad \tilde{\beta}_{d,t} \;=\; \beta_{d,t} - \beta_{d_{\max},t}$$

- i = pixel centroid; t = year (2017–2025)
- y_{it} = propensity score from **Google Satellite Embedding** for pixel i in year t
  - probit model fitted on 2020 embeddings, applied to all years
- α_i = pixel fixed effect; year 2020 omitted for identification ($\beta_{d,2020} \equiv 0$)
- bin_i = signed 100 m distance bin (negative = inside project boundary)
- $\tilde{\beta}_{d,t}$: re-centred so each year's line passes $y = 0$ at the outermost bin ($d_{\max}$)
- Standard errors clustered by 100 m distance bin



Propensity score — Spatial gradient by year
Year
2017
2021
2025
Coefficient (relative to 1000–1100 m bin)
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−200
0
200
400
600
800
1000
Distance to project boundary (m; negative = inside)